

- 1.1 Features of Open Source Operating Systems, Core Linux Distributions, Architecture, OS Services, System Calls, Run Levels.
- 1.2 File System: Hierarchical File System, File System features, Data Structures.
- 1.3 Process: Process concepts, context of process, Context Switch, Process State, State Transition diagram, Data Structure for processes.
- 1.4 Shell: Login into the system, Concept of Shell, Various Linux Shell and their Features.

Introduction to Linux Operating System

- Linux is a community of open-source Unix like operating systems that are based on the Linux Kernel.
- It was initially released by Linus Torvalds on September 17, 1991.
- It is a free and open-source operating system and the source code can be modified and distributed to anyone commercially or non-commercially under the GNU General Public License.
- Initially, Linux was created for personal computers and gradually it was used in other machines like servers, mainframe computers, supercomputers, etc. Nowadays, Linux is also used in embedded systems like routers, automation controls, televisions, digital video recorders, video game consoles, smartwatches, etc.
- The biggest success of Linux is Android (2005, operating system Google released in 2008) it is based on the Linux kernel that is running on smartphones and tablets.
- Due to android Linux has the largest installed base of all general-purpose operating systems. Linux is generally packaged in a Linux distribution.

Basic Features POMM HSS

Following are some of the important features of Linux Operating System.

- **Portable** – Portability means software can work on different types of hardware in the same way. Linux kernel and application programs support their installation on any kind of hardware platform.
- **Open Source** – Linux source code is freely available and it is a community based development project. Multiple teams work in collaboration to enhance the capability of Linux operating system and it is continuously evolving.
- **Multi-User** – Linux is a multiuser system means multiple users can access system resources like memory/ ram/ application programs at the same time.
- **Multiprogramming** – Linux is a multiprogramming system means multiple applications can run at the same time.
- **Hierarchical File System** – Linux provides a standard file structure in which system files/ user files are arranged.
- **Shell** – Linux provides a special interpreter program which can be used to execute commands of the operating system. It can be used to do various types of operations, call application programs. etc.
- **Security** – Linux provides user security using authentication features like password protection/ controlled access to specific files/ encryption of data.

Linux Distribution

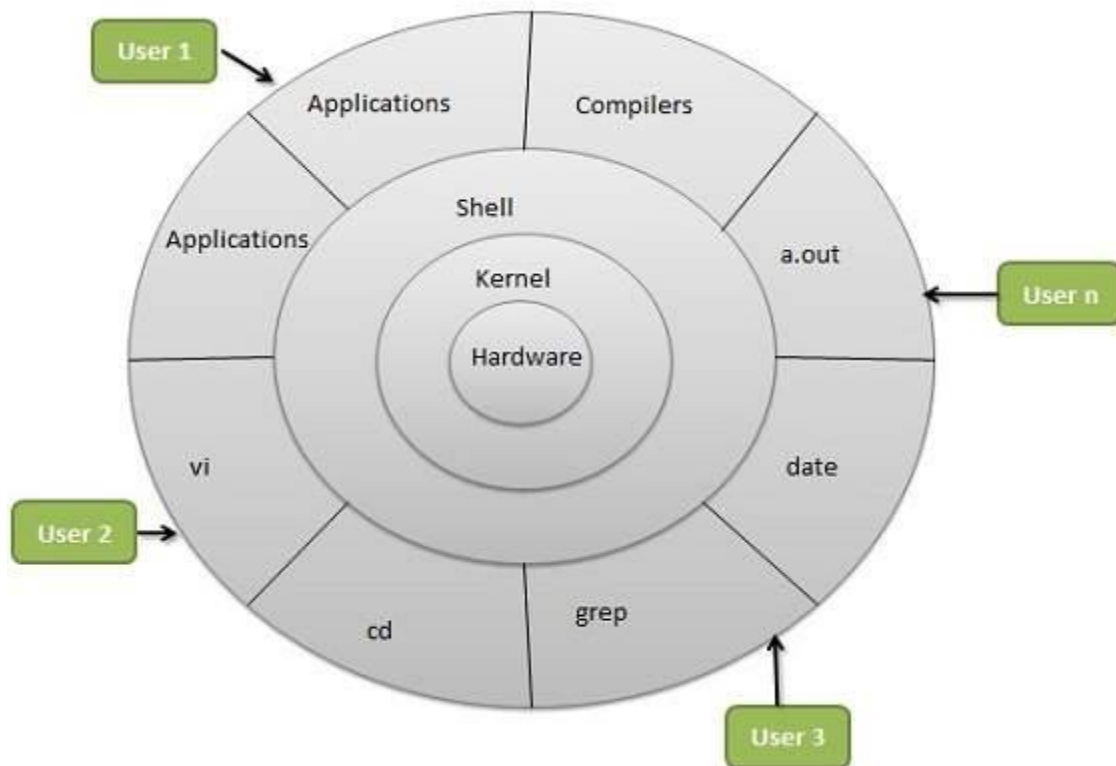
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Linux distribution is an operating system that is made up of a collection of software based on Linux kernel or a distribution contains the Linux kernel, supporting libraries and software. Linux operating system (any Linux distribution) can be downloaded and these distributions are available for different types of devices like embedded devices, personal computers, etc. Around 600 + Linux Distributions are available and some of the popular Linux distributions are:

- MX Linux
- Manjaro
- Linux Mint
- elementary
- Ubuntu
- Debian
- Solus
- Fedora
- openSUSE
- Deepin

Architecture of Linux

Linux architecture has the following components:



The architecture of a Linux System consists of the following layers –

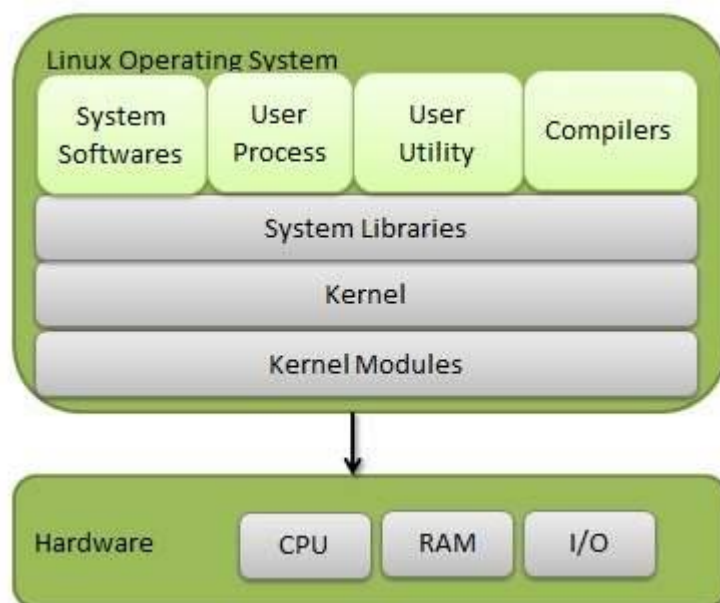
- **Hardware layer** – Hardware consists of all peripheral devices (RAM/ HDD/ CPU etc).
- **Kernel** – It is the core component of Operating System, interacts directly with hardware, provides low level services to upper layer components.
 - Different types of the kernel are:
 - Monolithic Kernel
 - Hybrid kernels
 - Exo kernels
 - Micro kernels

- **Shell** – An interface to kernel, hiding complexity of kernel's functions from users. The shell takes commands from the user and executes kernel's functions.
- **Utilities** – Utility programs that provide the user most of the functionalities of an operating systems.

Components of Linux System

Linux Operating System has primarily three components

- **Kernel** – Kernel is the core part of Linux. It is responsible for all major activities of this operating system. It consists of various modules and it interacts directly with the underlying hardware. Kernel provides the required abstraction to hide low level hardware details to system or application programs.
- **System Library** – System libraries are special functions or programs using which application programs or system utilities accesses Kernel's features. These libraries implement most of the functionalities of the operating system and do not requires kernel module's code access rights.
- **System Utility** – System Utility programs are responsible to do specialized, individual level tasks.



Kernel Mode vs User Mode

Kernel component code executes in a special privileged mode called **kernel mode** with full access to all resources of the computer. This code represents a single process, executes in single address space and do not require any context switch and hence is very efficient and fast. Kernel runs each processes and provides system services to processes, provides protected access to hardware to processes.

Support code which is not required to run in kernel mode is in System Library. User programs and other system programs works in **User Mode** which has no access to system hardware and kernel code. User programs/ utilities use System libraries to access Kernel functions to get system's low level tasks.

Advantages of Linux

- The main advantage of Linux, is it is an **open-source** operating system. This means the source code is easily available for everyone and you are allowed to contribute, modify and distribute the code to anyone without any permissions.
- In terms of security, Linux is more **secure** than any other operating system. It does not mean that Linux is 100 percent secure it has some malware for it but is less vulnerable than any other operating system. So, it does not require any anti-virus software.
- The **software updates in Linux are easy and frequent**.
- **Various Linux distributions are available** so that you can use them according to your requirements or according to your taste.
- Linux is **freely available** to use on the internet.
- It has large **community support**.
- It provides **high stability**. It rarely slows down or freezes and there is no need to reboot it after a short time.
- It maintains the **privacy** of the user.
- The **performance** of the Linux system is much **higher** than other operating systems. It allows a large number of people to work at the same time and it handles them efficiently.
- It is network friendly.
- The **flexibility** of Linux is high. There is no need to install a complete Linux suit; you are allowed to **install only required components**.
- Linux is **compatible** with a large number of **file formats**.
- It is **fast and easy to install** from the web. It can also install on any hardware even on your old computer system.
- It performs all tasks properly even if it has limited space on the hard disk.

Disadvantages of Linux

- It is not very user-friendly. So, it may be confusing for beginners.
- It has small peripheral hardware drivers as compared to windows.

Is There Any Difference between Linux and Ubuntu?

The answer is YES. The main difference between Linux and Ubuntu is Linux is the family of open-source operating systems which is based on Linux kernel, whereas Ubuntu is a free open-source operating system and the Linux distribution which is based on Debian. Or in other words, Linux is the core system and Ubuntu is the distribution of Linux. Linux is developed by Linus Torvalds and released in 1991 and Ubuntu is developed by Canonical Ltd. and released in 2004.

Operating System Services: PRIC FE

An Operating System provides services to both the users and to the programs.

- It provides programs an environment to execute.
- It provides users the services to execute the programs in a convenient manner.

Following are a few common **services** provided by an operating system –

- Program execution
- I/O operations
- File System manipulation
- Communication
- Error Detection
- Resource Allocation
- Protection

Program execution

Operating systems handle many kinds of activities from user programs to system programs like printer spooler, name servers, file server, etc. Each of these activities is encapsulated as a process.

A process includes the complete execution context (code to execute, data to manipulate, registers, OS resources in use). Following are the major activities of an operating system with respect to program management –

- Loads a program into memory.
- Executes the program.
- Handles program's execution.
- Provides a mechanism for process synchronization.
- Provides a mechanism for process communication.
- Provides a mechanism for deadlock handling.

I/O Operation

An I/O subsystem comprises of I/O devices and their corresponding driver software. Drivers hide the peculiarities of specific hardware devices from the users.

An Operating System manages the communication between user and device drivers.

- I/O operation means read or write operation with any file or any specific I/O device.
- Operating system provides the access to the required I/O device when required.

File system manipulation

A file represents a collection of related information. A file system is normally organized into directories on storage media for easy navigation and usage. These directories may contain files and other directions. Following are the major activities of an operating system with respect to file management –

- Program needs to read a file or write a file.
- The operating system gives the permission to the program for operation on file.
- Permission varies from read-only, read-write, denied and so on.
- Operating System provides an interface to the user to create/delete files.
- Operating System provides an interface to the user to create/delete directories.
- Operating System provides an interface to create the backup of file system.

Communication

In case of distributed systems there is need to share memory, peripheral devices, or a clock, the operating system manages communications between all the processes. Multiple processes communicate with one another through communication lines in the network.

The OS handles routing and connection strategies, and the problems of contention and security. Following are the major activities of an operating system with respect to communication –

- Two processes often require data to be transferred between them
- Both the processes can be on one computer or on different computers, but are connected through a computer network.
- Communication may be implemented by two methods, either by Shared Memory or by Message Passing.

Error handling

Errors can occur anytime and anywhere. An error may occur in CPU, in I/O devices or in the memory hardware. Following are the major activities of an operating system with respect to error handling –

- The OS constantly checks for possible errors.
- The OS takes an appropriate action to ensure correct and consistent computing.

Resource Management

In case of multi-user or multi-tasking environment, resources such as main memory, CPU cycles and files storage are to be allocated to each user or job. Following are the major activities of an operating system with respect to resource management –

- The OS manages all kinds of resources using schedulers.
- CPU scheduling algorithms are used for better utilization of CPU.

Protection

Considering a computer system having multiple users and concurrent execution of multiple processes, the various processes must be protected from each other's activities.

Protection refers to a mechanism or a way to control the access of programs, processes, or users to the resources defined by a computer system. Following are the major activities of an operating system with respect to protection –

- The OS ensures that all access to system resources is controlled.
- The OS ensures that external I/O devices are protected from invalid access attempts.
- The OS provides authentication features for each user by means of passwords.

System Calls

Run Levels in Linux

A run level is a state of init and the whole system that defines what system services are operating. Run levels are identified by numbers. Some system administrators use run levels to define which subsystems are working, e.g., whether X is running, whether the network is operational, and so on.

- Whenever a LINUX system boots, firstly the **init** process is started which is actually responsible for running other start scripts which mainly involves initialization of you hardware, bringing up the network, starting the graphical interface.
- Now, the **init** first finds the default **runlevel** of the system so that it could run the start scripts corresponding to the default run level.
- A **runlevel** can simply be thought of as the state your system enters like if a system is in a single-user mode it will have a **runlevel 1** while if the system is in a multi-user mode it will have a **runlevel 5**.
- A **runlevel** in other words can be defined as a **preset single digit integer** for defining the operating state of your LINUX or UNIX-based operating system. Each runlevel designates a different system configuration and allows access to different combination of **processes**.

The important thing to note here is that there are differences in the runlevels according to the operating system. The standard **LINUX kernel** supports these seven different runlevels :

Run Level	Mode	Action
0	Halt	Shuts down system
1	Single-User Mode	Does not configure network interfaces, start daemons, or allow non-root logins
2	Multi-User Mode	Does not configure network interfaces or start daemons.
3	Multi-User Mode with Networking	Starts the system normally.
4	Undefined	Not used/User-definable
5	X11	As runlevel 3 + display manager(X)
6	Reboot	Reboots the system

- 0 – System halt *i.e* the system can be safely powered off with no activity.
- 1 – Single user mode.
- 2 – Multiple user mode with no NFS(network file system).
- 3 – Multiple user mode under the command line interface and not under the graphical user interface.
- 4 – User-definable.
- 5 – Multiple user mode under GUI (graphical user interface) and this is the standard runlevel for most of the LINUX based systems.
- 6 – Reboot which is used to restart the system.

By default most of the LINUX based system boots to runlevel 3 or runlevel 5. In addition to the standard runlevels, users can modify the preset runlevels or even create new ones according to the requirement. Runlevels 2 and 4 are used for user defined runlevels and runlevel 0 and 6 are used for halting and rebooting the system.

Obviously the start scripts for each run level will be different performing different tasks. These start scripts corresponding to each run level can be found in special files present under **rc sub directories**.

At **/etc/rc.d** directory there will be either a set of files named **rc.0**, **rc.1**, **rc.2**, **rc.3**, **rc.4**, **rc.5** and **rc.6**, or a set of directories named **rc0.d**, **rc1.d**, **rc2.d**, **rc3.d**, **rc4.d**, **rc5.d** and **rc6.d**.

For example, run level 1 will have its start script either in file **/etc/rc.d/rc.1** or any files in the directory **/etc/rc.d/rc1.d**.

Changing runlevel

init is the program responsible for altering the run level which can be called using **telinit** command. "The **telinit** command (really just a symbolic link to **init**) enables you to specify a desired run level, causing the termination of all system processes that shouldn't exist in that run level, and starting all processes that should be running."

For example, to change a runlevel from 3 to runlevel 5 which will actually allow the GUI to be started in multi-user mode the **telinit** command can be used as :

/*using telinit to change

runlevel from 3 to 5*/

telinit 5

NOTE : The changing of runlevels is a task for the super user and not the normal user that's why it is necessary to be logged in as super user for the successful execution of the above telinit command or you can use **sudo** command as :

// using sudo to execute telinit

sudo telinit 5

The default runlevel for a system is specified in **/etc/inittab** file which will have an entry **id : 5 : initdefault** if the default runlevel is set to 5 or will have an entry **id : 3 : initdefault** if the default runlevel is set to 3.

Need for changing the runlevel

- There can be a situation when you may find trouble in **logging in** in case you don't remember the password or because of the corrupted **/etc/passwd** file (have all the user names and passwords), in this case the problem can be solved by booting into a single user mode *i.e* runlevel 1.
- You can easily halt the system by changing the runlevel to 0 by using

telinit 0.

RUN-LEVEL IN LINUX

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- 0 – halt (shutdown pc)
- 1 – Single user mode
- 2 – Multiuser
- 3 – Full multiuser mode
- 4 – unused
- 5 – X11 (Graphical)
- 6 – reboot

TO VIEW RUNLEVEL CONFIG FILE

```
# cat /etc/inittab
```

TO CHANGE RUNLEVEL CONFIGURATION FILE

```
# vi /etc/inittab
```

To check Current Run Level

```
# who -r      or
```

```
# runlevel
```

To change Run Level

```
# init 1
```

On most Linux server system default run level is 3 and on most Linux Desktop system default run level is 5.